

Project CldFound01

AWS Academy Cloud Foundations Modules 1 to 10

<https://awsacademy.instructure.com/>

First Name: **SANJAY**

Last Name: **DEVANG**



Figure 1

URL :- <https://www.credly.com/go/1hNRIIDx>

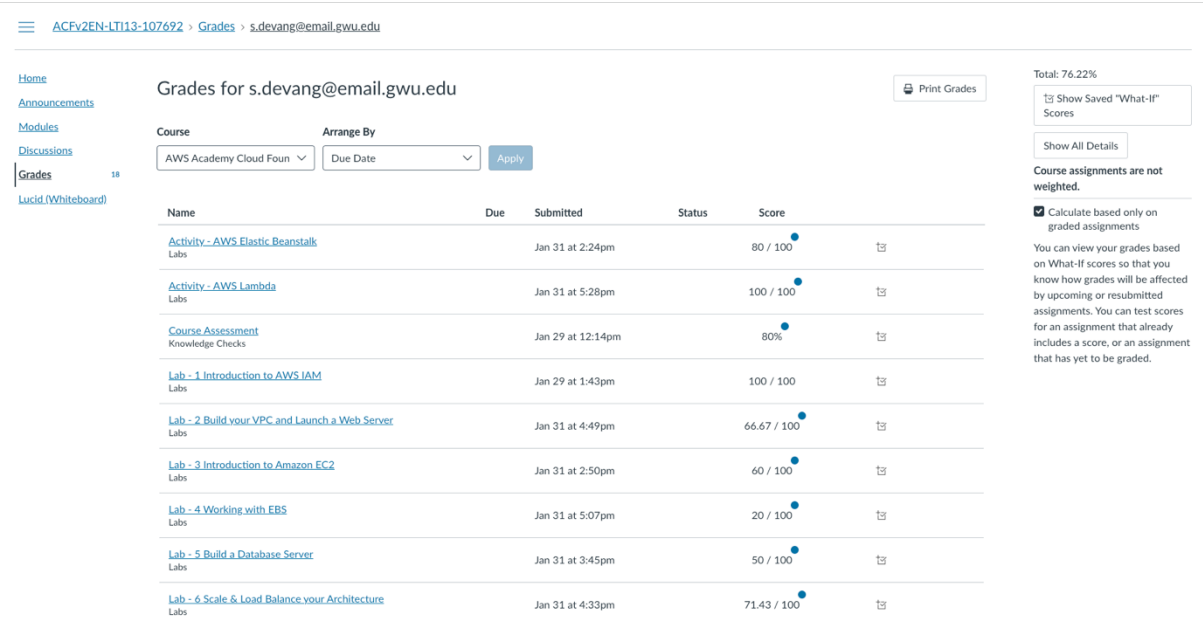


Figure 1: Knowledge Check and Lab Result

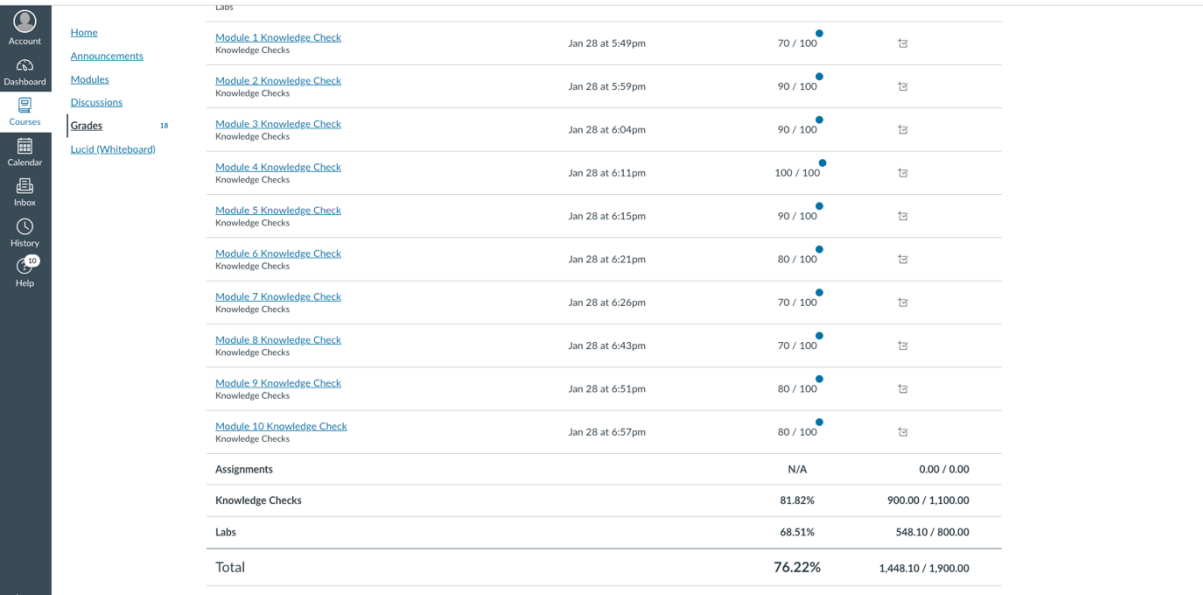


Figure 2 : Knowledge Check and Lab Result

Module 1 - Cloud Concepts Overview:

Introduction to Cloud Computing: I learned how cloud computing allows businesses to access on-demand IT resources over the internet, such as storage, databases, and servers, eliminating the need to maintain physical infrastructure (Amazon Web Services, n.d.-a).

Advantages of the Cloud: I understood the key benefits of using cloud services, including scalability, cost efficiency, and reliability, which help organizations to manage workloads more effectively and innovate rapidly (Amazon Web Services, n.d.-b).

Module 2 - Cloud Economics and Billing :

Fundamentals of Pricing: I discovered that AWS pricing structures, such as pay-as-you-go pricing, are made to offer flexibility and cost savings by enabling businesses to only pay for the resources they use up front.

AWS Billing and Cost Optimization: I learned about tools such as the AWS Billing Dashboard and how businesses can use features like budget alerts and comprehensive use statistics to track and optimize their cloud expenses.

Module 3 - AWS Global Infrastructure Overview :

AWS Global architecture: I learnt that AWS is built with a globally distributed architecture that includes Availability Zones, Regions, and Edge Locations to guarantee fault tolerance, low latency, and high availability for users everywhere.

AWS Services: In order to create scalable and reliable applications, I investigated how AWS provides a variety of services like computation, storage, networking, and databases that can be easily linked with the worldwide infrastructure.

Module 4 - AWS Cloud Security:

AWS Shared Responsibility Model: I learned how AWS security operates under a shared responsibility model where AWS secures the infrastructure, while customers are responsible for securing their data, identity access, and application configurations.

AWS Identity and Access Management (IAM): This module emphasized the importance of IAM, including how to create and manage roles, users, and policies to securely control access to AWS resources and maintain strong identity protection practices.

Module 5 - Networking and Content Delivery:

Amazon Virtual Private Cloud (VPC): I discovered that VPC makes it possible to establish logically isolated networks within AWS, enabling the setting up of NAT gateways, route tables, and public and private subnets to safely regulate traffic flow.

Amazon Route 53 and CloudFront: The important role of CloudFront for providing web content with low latency and high transfer speeds worldwide and Route 53 for managing domain names were highlighted in this module. For cloud applications to be scalable and effective, these services are essential.

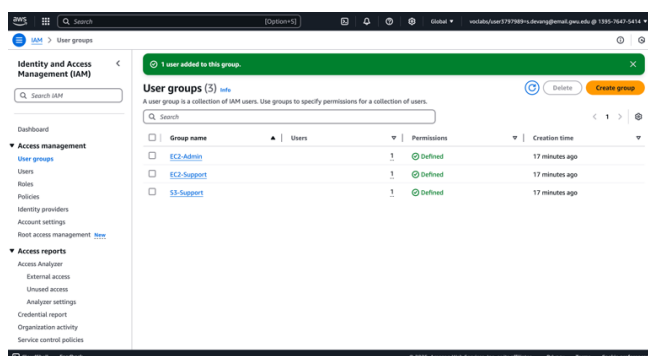


Figure 3 : AWS EC2 Instance Summary

Note :- The list of user groups in the Identity and Access Management (IAM) console is shown in this figure. Three user groups with defined permissions are displayed: EC2-Admin, EC2-Support, and S3-Support. By assembling people with particular AWS resource permissions, these groups control access.

Module 6 - Compute:

Overview of Amazon EC2: I learned that Amazon EC2 offers cloud computing capability that can be resized, enabling enterprises to build scalable applications with a variety of instance types, storage, and networking configurations to select from.

AWS Lambda: This module focused on the serverless architecture of AWS Lambda, which allows code to run without the need to setup or manage servers. Lambda is perfect for event-driven applications since it manages scaling and resources automatically.

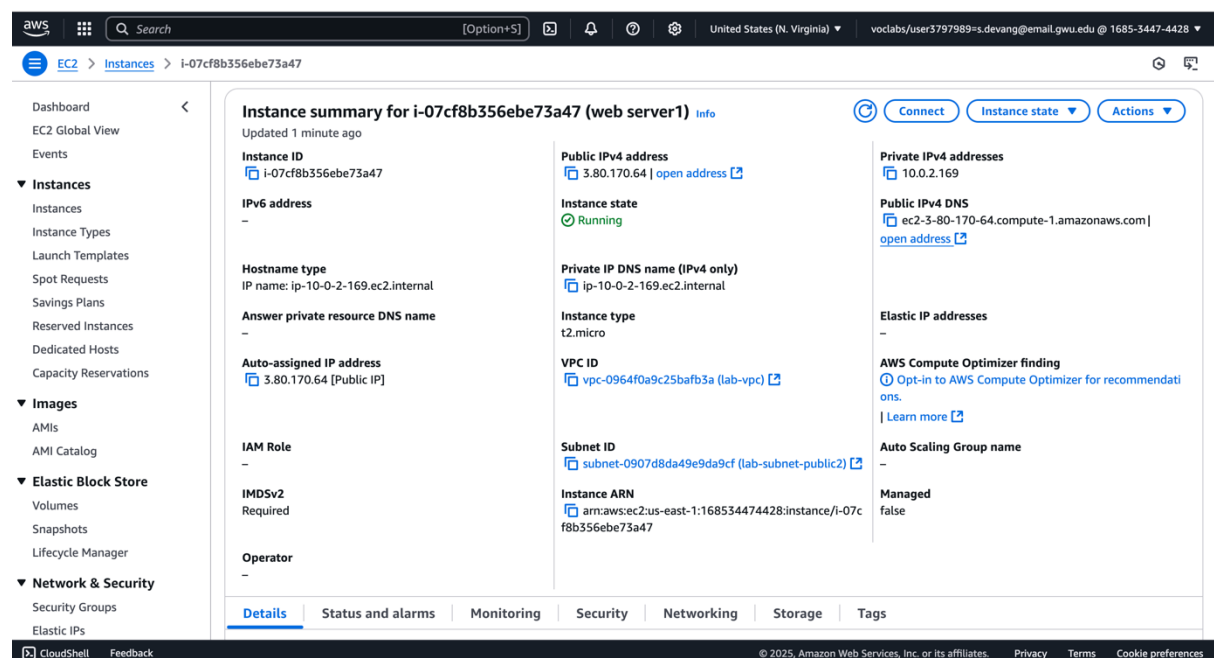


Figure 4 :Summary of an EC2 instance :

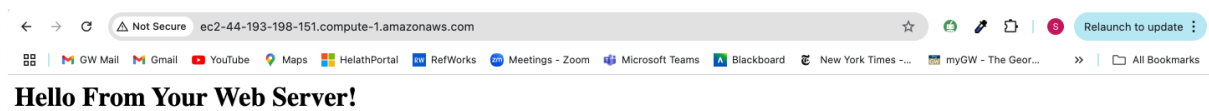


Figure 5: Output of the web server

Note :- The screenshot below displays an EC2 instance summary with the name "Web Server 1." It contains important details including the instance's public IPv4 address (3.80.170.64), private IP (10.0.2.169), instance state (Running), VPC, and subnet information. This shows that the instance is operational and reachable by the general public using the public IP address that was assigned to it.

 A screenshot of the AWS Management Console 'Instances' page. It shows a list of two EC2 instances, both in a 'Running' state. The page includes search filters, a table with instance details, and action buttons.

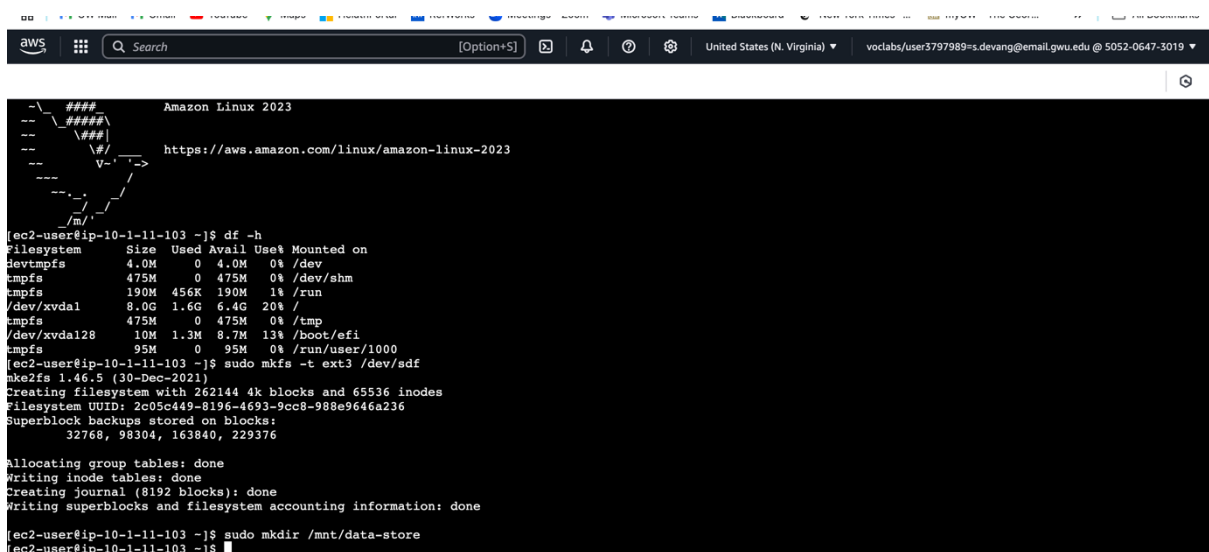
	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Av
☐	c1484-samp-uetPsoAruBr	i-Oe3e115b4b760284c	Running	t3.micro	3/3 checks passed	View alarms +	us-
☐	c1484-samp-uetPsoAruBr	i-Ofee4f246af1d657	Running	t3.micro	3/3 checks passed	View alarms +	us-

Figure 6 : AWS Elastic Beanstalk Environment Overview

Note :- The Elastic Beanstalk dashboard with deployed environments should be displayed in this figure. Details like the application version, instance metrics, health status, and environment name are usually included. Elastic Beanstalk automates resource provisioning, scaling, and monitoring, making web application setup and management easier.

Module 7 - Storage Overview:

I learned about Amazon EBS, Amazon S3, and Amazon EFS, among other AWS storage services. For EC2 instances, EBS offers block storage, allowing for low-latency, long-lasting storage. S3 provides high availability, scalable object storage that can handle a variety of data kinds. With elastic scaling, EFS enables shared file storage across several instances. In addition, I looked at Amazon S3 Glacier for affordable, safe archival storage and discovered methods for effectively managing data and maximizing storage expenses.



```

Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023

[ec2-user@ip-10-1-11-103 ~]$ df -h
Filesystem      Size  Used Avail Use% Mounted on
devtmpfs        4.0M   0  4.0M   0% /dev
tmpfs           475M   0  475M   0% /dev/shm
tmpfs           190M  456K  190M   1% /run
/dev/xvda1      8.0G  1.6G  6.4G  20% /
tmpfs           475M   0  475M   0% /tmp
/dev/xvda128    10M   1.3M   8.7M  13% /boot/efi
tmpfs           95M   0   95M   0% /run/user/1000
[ec2-user@ip-10-1-11-103 ~]$ sudo mkfs -t ext3 /dev/sdf
mkfs2fs 1.46.5 (30-Dec-2021)
Creating filesystem with 262144 4k blocks and 65536 inodes
filesystem UUID: 2c05c449-8196-4693-9cc8-988e9646a236
Superblock backups stored on blocks:
    32768, 98304, 163840, 229376

Allocating group tables: done
Writing inode tables: done
Creating journal (8192 blocks): done
Writing superblocks and filesystem accounting information: done

[ec2-user@ip-10-1-11-103 ~]$ sudo mkdir /mnt/data-store
[ec2-user@ip-10-1-11-103 ~]$

```

Figure 7 : Amazon Linux EC2 Instance with EBS Volume Mounted

Note:

This image shows the steps performed on an Amazon Linux EC2 instance to add and configure a new EBS volume. The following commands were executed:

- `df -h` displayed the current storage setup before mounting the new volume.
- `sudo mkfs -t ext3 /dev/sdf` created an ext3 file system on the attached EBS volume.
- `sudo mkdir /mnt/data-store` created a directory to mount the new volume.

- `sudo mount /dev/sdf /mnt/data-store` mounted the new volume under `/mnt/data-store`.

Finally, the new storage was verified using `df -h`, confirming the mounted volume is now part of the instance's available storage.

Module 8 - Databases Overview

□ **Amazon RDS:** I learned how RDS automates tasks like backups, scaling, and maintenance for relational databases, improving efficiency and reliability.

□ **Amazon DynamoDB:** I explored how DynamoDB handles highly scalable NoSQL data needs, supporting both key-value and document data models.

□ **Amazon Redshift:** I gained an understanding of how Redshift is optimized for large-scale data warehousing and analytics, processing petabytes of data efficiently.

□ **Amazon Aurora:** I discovered how Aurora offers high availability and performance with compatibility for MySQL and PostgreSQL, ideal for enterprise applications.

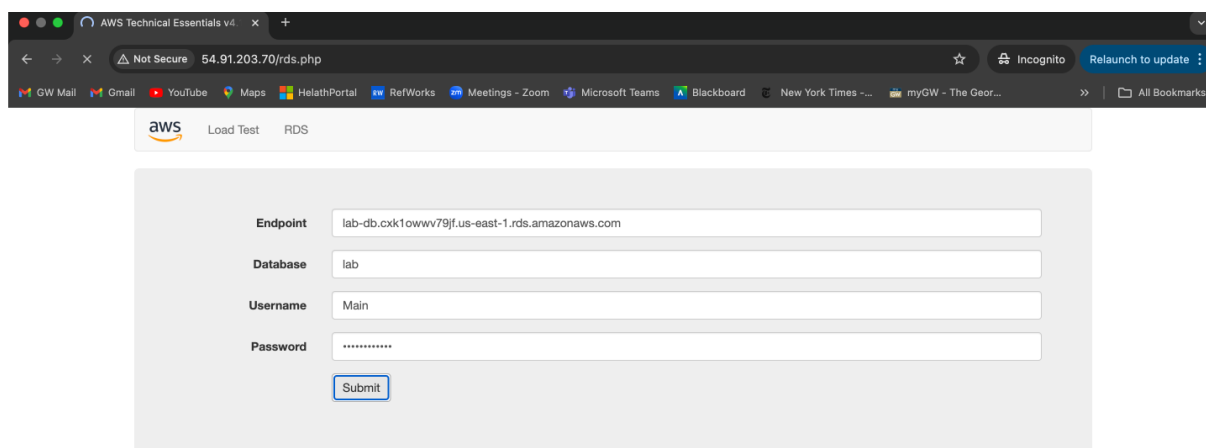


Figure 8: Connecting Web Application to Amazon RDS Database

Note: This Figure shows how a web application was successfully configured to communicate with an

Amazon RDS MySQL database. Values from the previously constructed RDS instance were entered in the endpoint, database name, username, and password boxes. By sending in these details, the connection was made, enabling fault-tolerant multi-AZ data storage and retrieval. This lab taught me how to set up subnet groups and security groups, start an RDS instance, and connect a web application to the database for persistent data management.

Module 9 - Cloud Architecture:

Cloud Architecture: Using the Well-Architected Framework, AWS highlights a methodical approach to creating scalable and robust cloud systems. Operational excellence, security, reliability, performance efficiency, and cost optimization are the five main pillars around which this framework is based.

Automating procedures and keeping an eye on system performance are key components of operational excellence, which guarantees ongoing enhancement and effective service provision. Implementing secure access restrictions, data protection, and redundancy is the main goal of security and reliability in order to prevent systems from malfunctions or data breaches. In order to manage fluctuating workloads, performance efficiency emphasizes the necessity of dynamic scaling and resource optimization. Finally, cost optimization emphasizes cutting costs through efficient resource management and fine-tuning.

This framework acts as a useful manual for creating reliable, secure, and effective cloud infrastructures that guarantee businesses the best possible service performance and dependability.

Module 10 - Auto Scaling and Monitoring

I learned how AWS features like EC2 Auto Scaling, Amazon CloudWatch, and Elastic Load Balancing

(ELB) guarantee system scalability and dependability. To preserve peak performance and avoid overload, ELB divides incoming traffic among several resources. Through metrics and alarms, CloudWatch provides actionable insights into resource utilization, application performance, and logs. EC2 Auto Scaling optimizes both performance and costs by dynamically adjusting the number of instances to maintain high availability. When used in tandem, these tools support scalable, robust systems under a range of workload conditions.

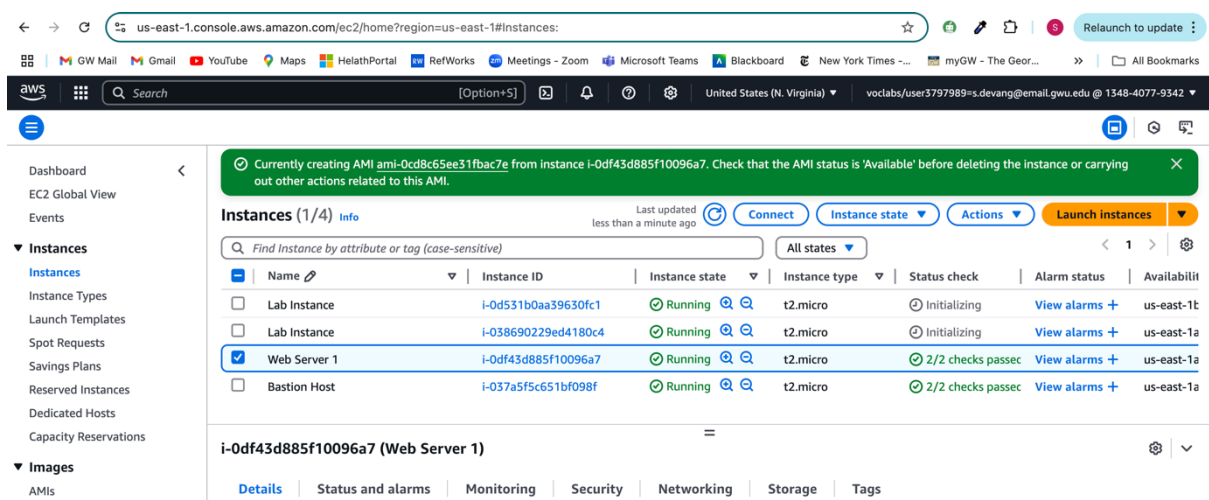


Figure 9 : Auto Scaling

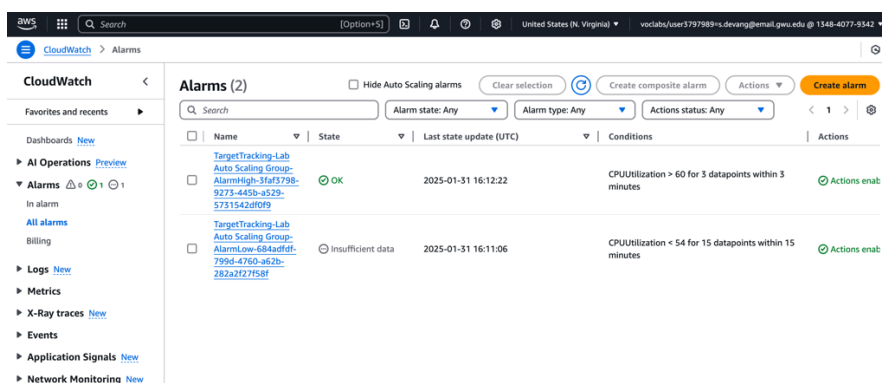


Figure 10 : CloudWatch Monitoring for Web Server Instances

Note: AWS Auto Scaling and CloudWatch monitoring are implemented in these screenshots. A

new AMI is being built while several EC2 instances are operating under auto-scaling in the original image. CloudWatch alarms set up to track CPU usage for scaling activities are seen in the second image. I gained knowledge about how CloudWatch monitors system performance and sets off alarms that dynamically modify the number of instances to preserve efficiency and performance, as well as how Elastic Load Balancing divides traffic among several instances.

References:

Amazon Web Services. (n.d.-a). *What is cloud computing?* AWS Documentation. Retrieved from <https://docs.aws.amazon.com/whitepapers/latest/aws-overview/what-is-cloud-computing.html>

Amazon Web Services. (n.d.-b). *Six advantages of cloud computing.* AWS Documentation. Retrieved from <https://docs.aws.amazon.com/whitepapers/latest/aws-overview/six-advantages-of-cloud-computing.html>